

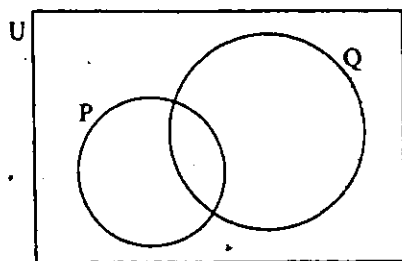
1. The number 3.14063 written correct to 3 decimal places is
- (A) 3.140
(B) 3.141
(C) 3.146
(D) 3.150
2. If $4.3 \times 0.37 = 1.591$, then 0.43×370 is
- (A) 1.591
(B) 15.91
(C) 159.1
(D) 1591.0
3. $5\frac{2}{5} + 4\frac{3}{7} =$
- (A) $9\frac{5}{35}$
(B) $9\frac{6}{35}$
(C) $9\frac{12}{35}$
(D) $9\frac{29}{35}$
4. Written correct to two significant figures, 18.96×2.03 is
- (A) 0.38
(B) 38
(C) 38.10
(D) 38.11
5. If \$350 is shared in the ratio 2:5, the SMALLER share is
- (A) \$250
(B) \$175
(C) \$100
(D) \$ 70
6. In a school the ratio of the number of pupils to the number of teachers is 20 : 1. If the number of pupils is 840, how many teachers are there?
- (A) 40
(B) 42
(C) 800
(D) 840
7. If $3n$ is an odd number, which of the following is an even number?
- (A) $3n - 1$
(B) $3n + 2$
(C) $3n - 2$
(D) $3n + 2n$
8. What is the HIGHEST common factor of the set of numbers {54, 72, 90}?
- (A) 9
(B) 18
(C) 90
(D) 1080
9. 99×101 has the same value as
- (A) $(99 \times 100) + 1$
(B) $(99 \times 100) - (99 \times 1)$
(C) $(99 \times 100) + (99 \times 1)$
(D) $(99 \times 100)(99 \times 1)$
10. The first three common multiples of 3, 4 and 6 are
- (A) 0, 1, 2
(B) 3, 4, 6
(C) 6, 8, 12
(D) 12, 24, 36

11. If $P = \{2, 3, 5, 7\}$,
 $Q = \{2, 3, 6\}$ and
 $S = \{2, 4, 5\}$

Then $P \cap Q \cap S =$

- (A) $\{2\}$
 (B) $\{2, 3\}$
 (C) $\{2, 3, 5\}$
 (D) $\{2, 3, 4, 5, 6, 7\}$

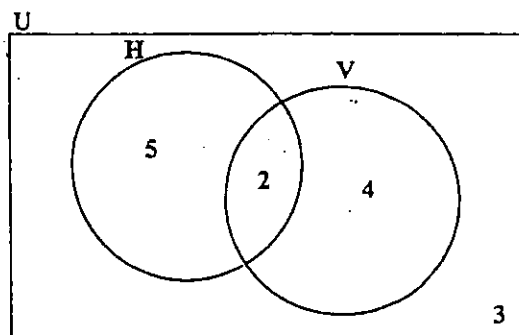
Item 12 refers to the Venn diagram below.



In the Venn diagram, $n(P) = 5$, $n(Q) = 9$
 and $n(P \cup Q) = 10$.

12. What is $n(Q \cap P)$?
- (A) 1
 (B) 4
 (C) 5
 (D) 9

Item 13 refers to the Venn diagram below.



13. In the Venn diagram

$U = \{\text{students who play games}\}$
 $H = \{\text{students who play hockey}\}$
 $V = \{\text{students who play volleyball}\}$

The number of students in each set is shown. How many students do NOT play volleyball?

- (A) 2
 (B) 3
 (C) 5
 (D) 8
14. All students in a class play Scrabble or Checkers or both. If 36% of the students play Scrabble only and 48% of the students play Checkers only, what percentage of students play BOTH games?
- (A) 12
 (B) 16
 (C) 84
 (D) 88
15. The cost of a calculator exclusive of sales tax is \$120. Susan bought a calculator and had to pay a sales tax of 10%. How much change would she receive from \$140?
- (A) \$ 8.00
 (B) \$12.00
 (C) \$28.00
 (D) \$32.00

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16. The simple interest on \$600 for 4 years at 5 percent per annum is

- (A) $\$ \frac{600 \times 5 \times 4}{100}$
 (B) $\$ \frac{100 \times 5 \times 4}{600}$
 (C) $\$ \frac{600 \times 5}{100 \times 4}$
 (D) $\$ \frac{100 \times 4}{600 \times 5}$

17. A television set costs \$350 cash. When bought on hire-purchase, a deposit of \$35 is required, followed by 12 monthly payments of \$30. How much is saved by paying cash?

- (A) \$10
 (B) \$25
 (C) \$40
 (D) \$45

18. A plot of land is valued at \$18 000. Land tax is charged at the rate of \$0.70 per \$100. What is the TOTAL amount of tax paid for the land?

- (A) \$110.00
 (B) \$126.00
 (C) \$180.70
 (D) \$257.15

19. A loan of \$8 000 was paid back in 2 years in monthly payments of \$400.00. The interest on the loan, as a percentage, was

- (A) 5%
 (B) $8\frac{1}{3}\%$
 (C) $16\frac{2}{3}\%$
 (D) 20%

20. A store offers a discount of 10% to customers who spend more than \$20. If a customer's total bill was \$80, what will he actually pay?

- (A) \$60
 (B) \$70
 (C) \$72
 (D) \$74

Item 21 refers to the table below which shows the rates charged by an insurance company.

House Insurance	25¢ per \$100
Contents Insurance	50¢ per \$100

21. How much will a person pay for his insurance, if his house is valued at \$50 000, and the contents at \$10 000?

- (A) \$175
 (B) \$225
 (C) \$450
 (D) \$500

22. A car presently valued at \$12 000 depreciates at the rate of 10% per annum. What will be the value of the car one year later?

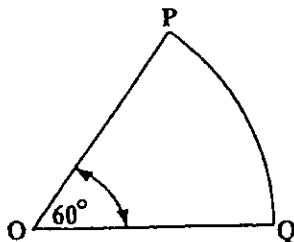
- (A) \$10 800
 (B) \$11 800
 (C) \$11 880
 (D) \$13 200

23. The product of a number p and its reciprocal may be written as

- (A) $p^2 \times \frac{1}{p}$
 (B) $p^2 \times -p$
 (C) $p \times \frac{1}{p}$
 (D) $p \times -p$

24. $5(2x - y) - 2(3y - 5x) =$
- (A) $20x - 11y$
 (B) $5x - 7y$
 (C) $2x - 6y$
 (D) $-11y$
25. The value of $(-2a)(-3a)$ is
- (A) $-6a$
 (B) $-5a$
 (C) $5a^2$
 (D) $6a^2$
26. If $50 - 3x = x - 26$, then $x =$
- (A) -12
 (B) -6
 (C) 6
 (D) 19
27. If $15^2 = 225$ then the square root of 0.0225 is
- (A) 0.015
 (B) 0.15
 (C) 1.5
 (D) 15.0
28. If $a = 5$, $b = 2$, $c = 4$, then $6ab - 2c =$
- (A) 5
 (B) 52
 (C) 60
 (D) 628
29. If x is an integer which satisfies the inequalities $4 < x - 2 < 8$, then the SMALLEST possible value of x is
- (A) 7
 (B) 6
 (C) 5
 (D) 4
30. If $m * n = \sqrt{m^2 - n^2}$, then $5 * 3 =$
- (A) 2
 (B) $\sqrt{8}$
 (C) 4
 (D) $\sqrt{34}$
31. $2^6 \div 2^2 =$
- (A) 2^3
 (B) 2^4
 (C) 2^8
 (D) 2^{12}
32. How many kilograms are there in one tonne?
- (A) 10
 (B) 100
 (C) 1 000
 (D) 10 000
33. If it took a speed-boat 9 hours to travel a distance of 1 080 km, what was its average speed, in kmh^{-1} ?
- (A) 12
 (B) 102
 (C) 120
 (D) 1200
34. The distance around a lake is 8 km. On a map, this distance around the lake is represented by a length of 2 cm. The scale on the map is
- (A) 1 : 40
 (B) 1 : 2 000
 (C) 1 : 200 000
 (D) 1 : 400 000

Item 35 refers to the diagram below, not drawn to scale, which shows the sector of a circle with centre, O.



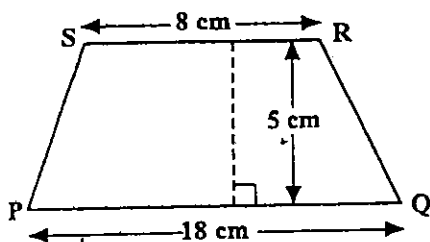
35. In the diagram, the length of the minor arc, PQ, is 8 cm. What is the circumference of the circle?

(A) 16 cm
(B) 24 cm
(C) 48 cm
(D) 64 cm

36. Fifty guests each had 2 glasses of champagne. Each glass held 150 millilitres. How many litres of champagne were used?

(A) 0.15
(B) 1.5
(C) 15
(D) 150

Item 37 refers to the diagram below, not drawn to scale, which shows a trapezium.



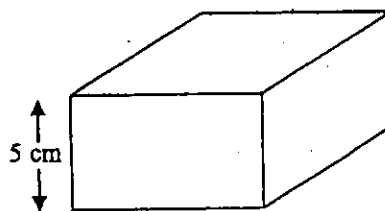
37. The area of the trapezium is

(A) 45 cm²
(B) 65 cm²
(C) 90 cm²
(D) 130 cm²

38. A boy leaves home at 09:15 hours and arrives at school at 10:05 hours. If he travels non-stop at an average speed of 6 kmh⁻¹, how many km is his home from school?

(A) 2 km
(B) 5 km
(C) 6 km
(D) 9 km

Item 39 refers to the diagram below, not drawn to scale, which shows a cuboid.



39. The volume of the cuboid is 320 cm³ and the height is 5 cm. If the cuboid has a square base, what is the length of one side of the base?

(A) 8 cm
(B) 16 cm
(C) 32 cm
(D) 64 cm

Item 40 refers to the information below.

The following scores were obtained by eleven footballers in a goal-shoot competition:

5, 3, 6, 8, 7, 8, 3, 11, 6, 3, 3

40. The modal score is

(A) 3
(B) 6
(C) 8
(D) 11

41. The table below shows the frequency of scores obtained by students in a test.

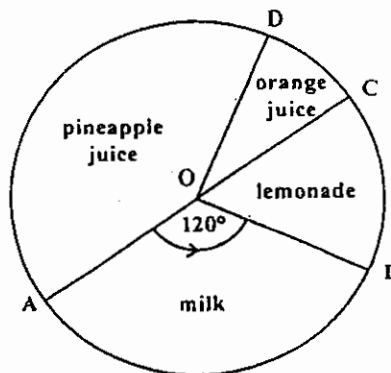
Scores	2	3	5	6	8	10
No. of Students	8	4	6	3	9	2

The range of the scores is

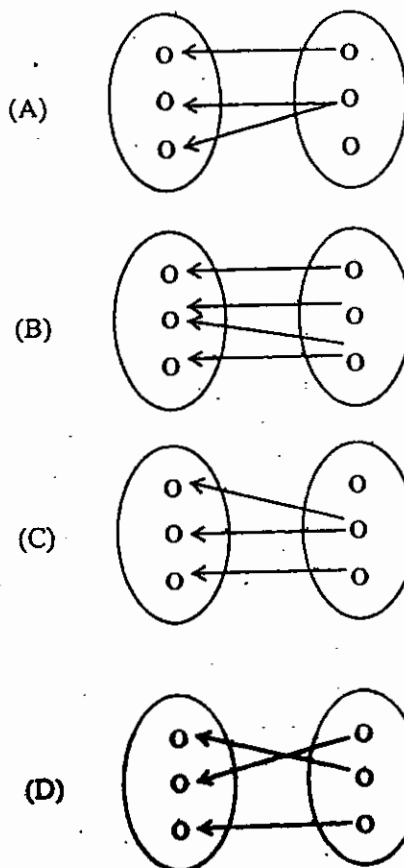
- (A) 2
(B) 7
(C) 8
(D) 10
42. Six hundred students write an examination. The probability of a randomly selected student failing the examination is $\frac{1}{5}$.
How many students are expected to pass?
- (A) 120
(B) 480
(C) 500
(D) 600
43. In a class of 20 students, 12 are girls. What is the probability that a student chosen at random is a boy?
- (A) $\frac{1}{2}$
(B) $\frac{2}{5}$
(C) $\frac{3}{5}$
(D) $\frac{2}{3}$
44. If the mean of the four numbers 4, 8, x and 12 is 10, then x is

- (A) 4
(B) 10
(C) 12
(D) 16

Item 45 refers to the pie-chart below which shows the preference in drinks of a group of students. AOC is a diameter of the circle.



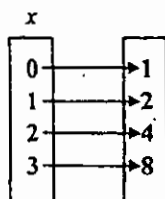
45. If 12 students prefer lemonade, then the TOTAL number of students is
- (A) 48
(B) 72
(C) 180
(D) 360
46. Which of the following diagrams BEST illustrates a function?



47. The equation of the line which passes through the point $(0, 2)$ and has a gradient of $\frac{1}{3}$ is

- (A) $y = 3x$
 (B) $y = 3x + 2$
 (C) $y = \frac{1}{3}x$
 (D) $y = \frac{1}{3}x + 2$

Item 48 refers to the diagram below.



48. The diagram represents the mapping

- (A) $x \rightarrow x + 2$
 (B) $x \rightarrow x + 1$
 (C) $x \rightarrow 2^x$
 (D) $x \rightarrow 2x$

49. If $f(x) = 2x^2 - 1$, then $f(-3) =$

- (A) -32
 (B) -19
 (C) 17
 (D) 35

50. Which of the following sets is represented by the function $f: x \rightarrow x^2 + 3$ where $x \in \{0, 1, 2, 3\}$?

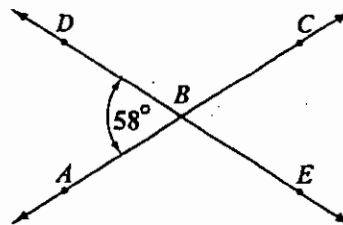
- (A) $\{(0, 3), (1, 4), (2, 7), (3, 12)\}$
 (B) $\{(0, 3), (1, 5), (2, 7), (3, 9)\}$
 (C) $\{(0, 3), (1, 4), (2, 5), (3, 6)\}$
 (D) $\{(0, 3), (1, 1), (2, 4), (3, 9)\}$

51. The range of $f: x \rightarrow x^2$ for the domain $\{-2, -1, 0, 1, 2\}$ is

- (A) $\{0, 1, 4\}$
 (B) $\{0, 2, 4\}$
 (C) $\{-4, -1, 0, 1, 4\}$
 (D) $\{-4, -2, 0, 2, 4\}$

Items 52–53 refer to the diagram below.

AC and DE are straight lines intersecting at B . Angle $DBA = 58^\circ$.



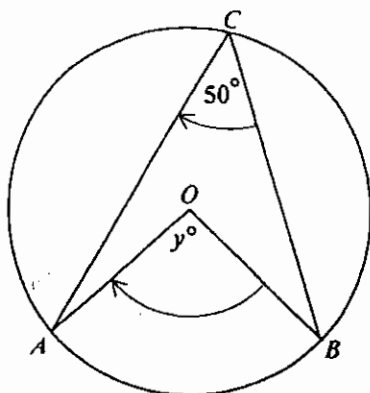
52. Which of the following angles are equal?

- (A) $\angle DBC$ and $\angle CBE$
 (B) $\angle CBE$ and $\angle ABE$
 (C) $\angle ABD$ and $\angle CBD$
 (D) $\angle ABD$ and $\angle CBE$

53. The measure of angle ABE is

- (A) 58°
 (B) 122°
 (C) 142°
 (D) 302°

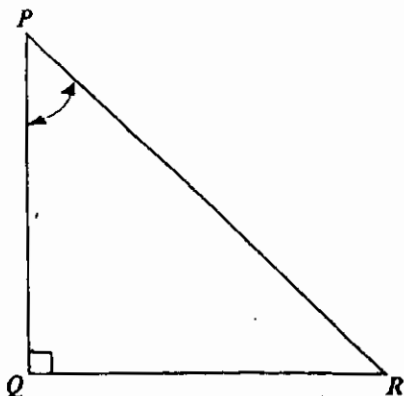
Item 54 refers to the diagram below.



54. If O is the centre of the circle, then y° is

- (A) 25°
- (B) 80°
- (C) 90°
- (D) 100°

Item 55 refers to the triangle PQR below.



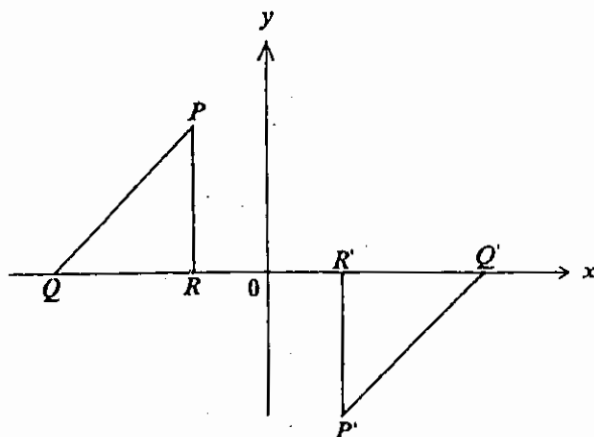
55. In the triangle $\cos P$ may be expressed as

- (A) $\frac{QR}{PR}$
- (B) $\frac{QR}{PQ}$
- (C) $\frac{PQ}{PR}$
- (D) $\frac{PQ}{QR}$

56. The image of a point $P(-2, 3)$ under a translation $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is

- (A) $(-6, 12)$
- (B) $(-5, -1)$
- (C) $(1, 7)$
- (D) $(5, 1)$

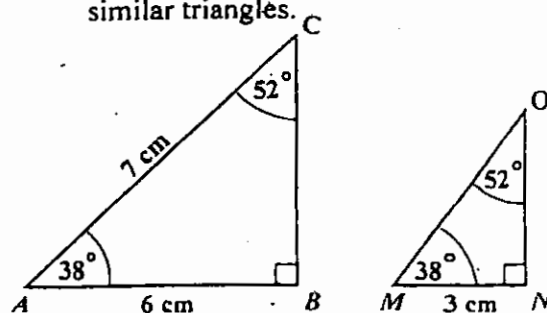
Item 57 refers to the diagram below which shows a transformation.



57. The transformation which will map triangle PQR onto $P'Q'R'$ is

- (A) a rotation of 180° about the origin
- (B) a reflection in the line $y = x$
- (C) an enlargement, centre R , of scale factor -1
- (D) a translation parallel to the x -axis

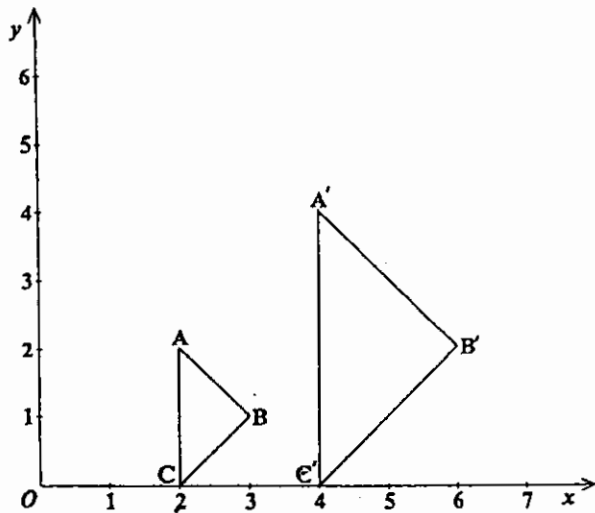
Item 58 refers to the following pair of similar triangles.



58. The length of MO , in centimetres, is

- (A) 7
- (B) 6
- (C) 3.5
- (D) 3

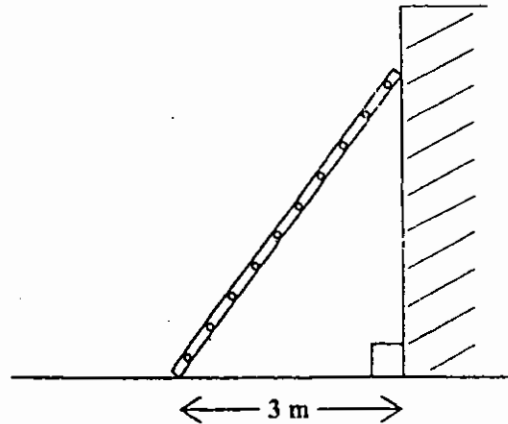
Item 59 refers to the diagram below which shows an enlargement.



59. In the diagram, triangle ABC is mapped onto triangle A'B'C' where C is the centre of enlargement. What is the scale factor of the enlargement?

- (A) $\frac{1}{2}$
- (B) -2
- (C) $-\frac{1}{2}$
- (D) 2

Item 60 refers to the diagram below which shows a ladder 5 metres long leaning against a vertical wall. The foot of the ladder is 3 metres away from the wall.



60. How far up the wall does the ladder reach?

- (A) 4 m
- (B) 6 m
- (C) 8 m
- (D) 15 m

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.